

3	(a)	loss in GPE = gain in elastic PE $mgh = \frac{1}{2}kx^2$ $m = \frac{kx^2}{2gh} = \frac{(25)(0.050)^2}{2(9.81)(0.110)} = 0.028959 \text{ kg}$ = 0.029 kg (shown)	M1
	(b)	When the spring first compresses, the magnitude of the <u>force from the spring is less than weight</u>. Hence, there is still a <u>downward resultant force</u> that causes the marble to continue accelerating.	B1 B1
	(c)	Max speed of marble happens when force from spring = weight $kx = mg$ $x = \frac{mg}{k} = \frac{(0.028959)(9.81)}{25} = 0.01136 \text{ m}$ $\text{gain in EPE} = \frac{1}{2}kx^2 = \frac{1}{2}(25)(0.01136)^2 = 0.001614 \text{ J}$ $\text{loss in GPE} = mg(0.060 + 0.01136)$ $= (0.028959)(9.81)(0.060 + 0.01136) = 0.02027 \text{ J}$ $\text{gain in KE} = \text{loss in GPE} - \text{gain in EPE}$ $= 0.02027 - 0.001614$ $= 0.019 \text{ J (2sf)}$	M1 M1 M1 A1

4	(a)	When a charged particle is travelling in a magnetic field, it	
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